Existence, Uniqueness, and Stability of Ground-State Solutions for the Nonlinear Schrödinger Equation in Optical Fibers

Eduard Kirr, PhD
Associate Professor
University of Illinois - Urbana-Champaign

Ethan Knox, BS
Master's Student
University of Illinois - Urbana-Champaign

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Abstract

We study multi-peak ground-state solutions of the nonlinear Schrödinger equation $(-\Delta + V + E)\,\psi + \sigma\,|\psi|^{2p}\,\psi = 0$ in the high-energy regime $E \to \infty$. Through the rescaling and concentration compactness framework of Kirr and Natarajan (2018)[10], ground states decompose into well-separated profiles whose positions are governed by a finite-dimensional algebraic system balancing the Hessian H_V of the potential against nonlinear inter-peak interactions.

We systematically classify all admissible multi-peak configurations near a non-degenerate critical point of V: collinear arrangements along a single eigenvector of H_V, and triangular configurations (isosceles, equilateral, and equilateral with rotational degree of freedom θ) in the plane spanned by two eigenvectors. For the rotational case, we derive explicit admissibility regions in (μ, θ) parameter space, where $\mu = \frac{|\lambda_2|}{|\lambda_1|}$ is the eigenvalue ratio: all orientations are admissible when $\frac{1}{3} \leq \mu \leq 3$, with restricted angular windows otherwise.

To extend these limiting configurations to exact solutions at finite energy, we formulate a perturbation system $F_i\,(\delta y, R) = 0$, construct explicit solutions at $R = 0$, and show that the linearization has a nontrivial kernel generated by the translational invariance of the interaction. Following the iterative kernel decomposition methodology of Kirr and Natarajan (who resolve analogous obstructions at the PDE and Lyapunov-Schmidt levels by modding out the symmetry responsible for each kernel) we decompose the reduced system into projections parallel and perpendicular to this kernel and identify the rotational equivariance as the structural property enabling application of the implicit function theorem on the complement.

1 Introduction

The propagation of intense light through optical fibers is governed by the interplay between linear dispersion and the nonlinear response of the dielectric medium. In silica, the centrosymmetric lattice eliminates second-order nonlinearity, so the leading nonlinear effect is the Kerr term $\chi^{(3)}$. Starting from Maxwell's equations in a source-free, non-magnetic dielectric, the electric field can be written as a slowly varying envelope modulating a carrier:

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{2}\hat{\mathbf{x}}\left[F(x,y)\,A(z,t)\,e^{i(\beta_0 z-\omega_0 t)} + \text{c.c.}\right].$$

Here $F(x,y)$ is the transverse modal shape set by the fiber's refractive-index profile, $A(z,t)$ is the envelope, and β_0 is the propagation constant at the carrier frequency ω_0. Applying the slowly varying envelope approximation and expanding the propagation constant to second order in frequency yields the nonlinear Schrödinger equation (NLS):

$$i\partial_z A - \frac{\beta_2}{2}\partial_T^2 A + \gamma\,|A|^2\,A = 0,$$

where β_2 is the group-velocity-dispersion parameter and $\gamma = \dfrac{n_2\omega_0}{cA_{\text{eff}}}$ encodes the Kerr nonlinearity.

In real fibers the refractive index is not spatially uniform. Fiber Bragg gratings impose periodic modulations, graded-index fibers create a parabolic transverse profile (an effective harmonic-oscillator potential), and photonic-crystal fibers achieve confinement via a micro-structured air-hole lattice. In every case, the spatial variation of the index enters the governing equations as an external potential $V(\mathbf{x})$. The transverse modal equation

$$\nabla_\perp^2 F + \left[\left(\frac{n\omega}{c}\right)^2(\mathbf{x}_\perp) - \beta^2\right]F = 0$$

is mathematically identical to the stationary Schrödinger equation $-\Delta\psi + V(x)\,\psi = E\psi$. Guided modes of the fiber correspond to bound states of this effective potential, and the fundamental mode is its ground state. This equivalence motivates the present work: engineering $V(\mathbf{x})$ via refractive-index design directly translates into questions about the existence, uniqueness, and stability of ground-state solutions of the NLS with an external potential.

Consequently, we study the stationary NLS

$$(-\Delta + V + E)\,\psi + \sigma\,|\psi|^{2p}\,\psi = 0, \tag{1}$$

with $V:\mathbb{R}^n\to\mathbb{R}$, $\psi:\mathbb{R}^n\to\mathbb{C}$, $\sigma>0$, and $E\geq 0$. The parameter E represents the energy (or, in the optical setting, the propagation constant). For each admissible E we seek ground-state solutions ψ_E and analyze how their spatial structure is governed by the geometry of V.

Because 1 is posed on the unbounded domain $\mathbb{R}^n$, compactness is lost and standard variational methods fail. The concentration compactness principle introduced by Lions[13] classifies minimizing sequences into compactness, vanishing, or dichotomy. By rescaling

$$u_E(x) = E^{-\frac{1}{2p}}\psi_E\left(E^{-\frac{1}{2}}x\right),$$

(see the global bifurcation analysis of Kirr and Natarajan[10]) one obtains a bounded family in H^1. Concentration compactness then yields a multi-peak decomposition: u_E is approximated by a finite sum of well-separated profiles, each solving the translation-invariant NLS $(-\Delta+1)\,u + \sigma\,|u|^{2p}\,u = 0$. The rescaled peak positions converge to the critical points of V as $E\to\infty$.

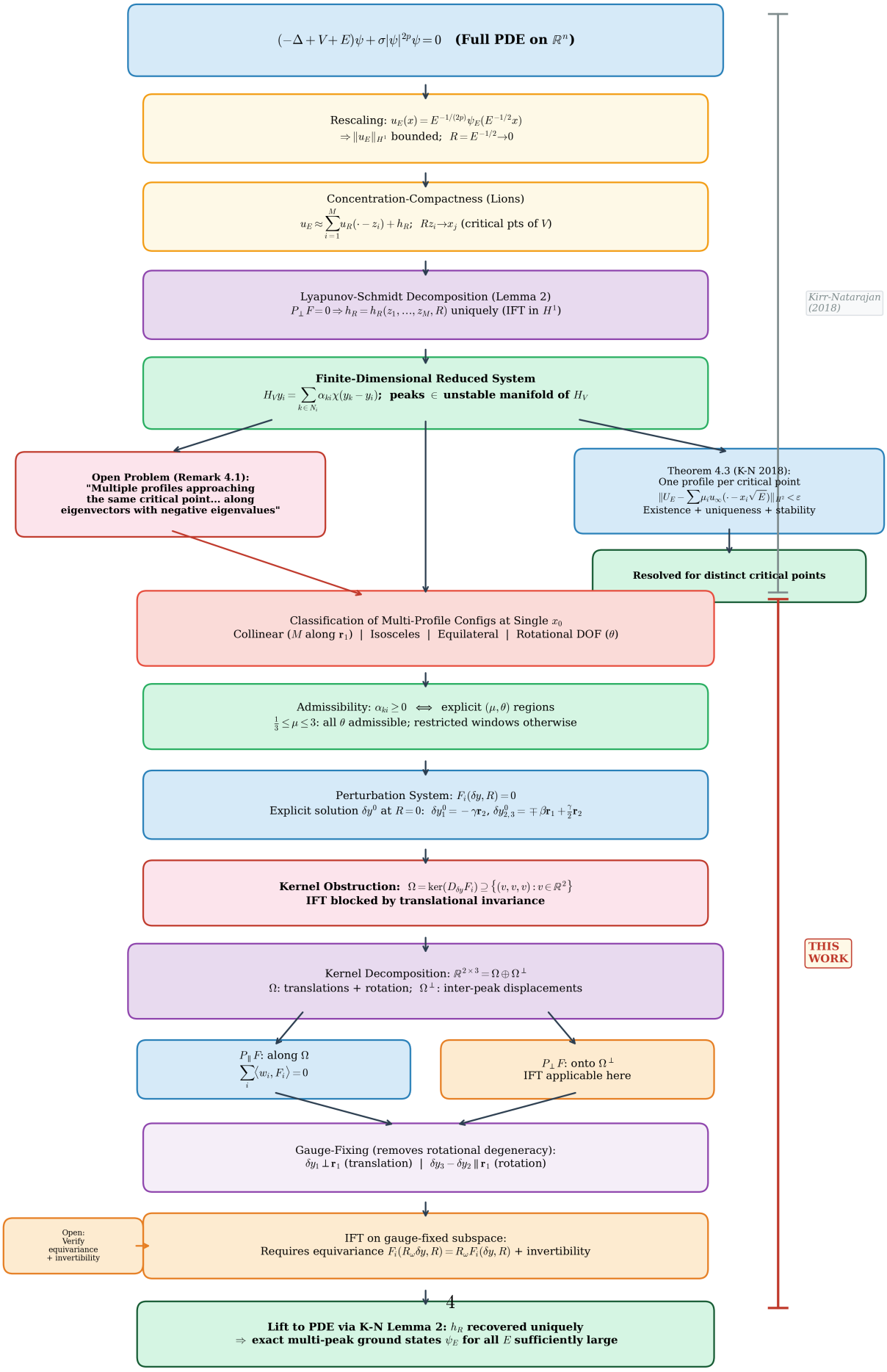
This multi-peak decomposition reduces the infinite-dimensional problem to a finite-dimensional system governing the peak locations. A Lyapunov-Schmidt reduction first solves for the orthogonal remainder and then projects onto the span of the translational modes. The resulting algebraic system balances the restoring force from the Hessian H_V of the potential against the attractive interaction between neighboring peaks:

$$H_V(x_0)\,y_i = \sum_{k\in N_i}\alpha_{ki}\chi\,(y_k - y_i),$$

where y_i are the rescaled peak positions, $\alpha_{ki}\in[0,1]$ are interaction coefficients, and $\chi>0$ quantifies the overall interaction strength. The peak set must lie in the unstable manifold of H_V.

We classify admissible multi-peak configurations (collinear, coplanar, and triangular) by analyzing the Hessian eigenvalues $\lambda_1, \lambda_2 < 0$ and the ratio $\mu = \frac{|\lambda_2|}{|\lambda_1|}$. The interaction constraints $\alpha_{ki} \geq 0$ restrict the admissible (μ, θ) parameter space, where θ is the orientation of a triangle relative to the eigenvector basis. For example, an equilateral triangle requires $\lambda_2 = 3\lambda_1$, while any triangle is admissible for $\frac{1}{3} \leq \mu \leq 3$.

Our contribution is a systematic analysis of this reduced system, yielding explicit geometric characterizations of multi-peak ground states near non-degenerate critical points of V. The analysis extends the concentration compactness and Lyapunov-Schmidt framework by handling the nontrivial kernel of the linearized reduced system. By decomposing the system into parallel and perpendicular components, exploiting rotational symmetry, and constructing solutions for finite scaling parameters $R = E^{-\frac{1}{2}}$, we obtain families of exact multi-peak solutions of 1 for all sufficiently large energies E.


$(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0$ **(Full PDE on $\mathbb{R}^n$)**
Rescaling: $u_E(x) = E^{-1/(2p)}\psi_E(E^{-1/2}x)$
$\Rightarrow \|u_E\|_{H^1}$ bounded; $R = E^{-1/2} \to 0$
Concentration-Compactness (Lions)
$u_E \approx \sum_{i=1}^{M} u_R(\cdot - z_i) + h_R$; $Rz_i \to x_j$ (critical pts of V)
Lyapunov-Schmidt Decomposition (Lemma 2)
$P_\perp F = 0 \Rightarrow h_R = h_R(z_1, \ldots, z_M, R)$ uniquely (IFT in H^1)
Kirr-Natarajan (2018)
Finite-Dimensional Reduced System
$H_V y_i = \sum_{k \in N_i} \alpha_{ki}\chi(y_k - y_i)$; **peaks $\in$ unstable manifold of H_V**
Open Problem (Remark 4.1): "Multiple profiles approaching the same critical point... along eigenvectors with negative eigenvalues"
Theorem 4.3 (K-N 2018): One profile per critical point
$\|U_E - \sum \mu_i u_\infty(\cdot - x_i\sqrt{E})\|_{H^2} < \varepsilon$
Existence + uniqueness + stability
Resolved for distinct critical points
Classification of Multi-Profile Configs at Single x_0
Collinear (M along $\mathbf{r}_1$) | Isosceles | Equilateral | Rotational DOF (θ)
Admissibility: $\alpha_{ki} \geq 0 \iff$ explicit (μ, θ) regions
$\frac{1}{3} \leq \mu \leq 3$: all θ admissible; restricted windows otherwise
Perturbation System: $F_i(\delta y, R) = 0$
Explicit solution δy^0 at $R = 0$: $\delta y_1^0 = -\gamma \mathbf{r}_2$, $\delta y_{2,3}^0 = \mp \beta \mathbf{r}_1 + \frac{\gamma}{2}\mathbf{r}_2$
Kernel Obstruction: $\Omega = \ker(D_{\delta y} F_i) \supseteq \{(v, v, v) : v \in \mathbb{R}^2\}$
IFT blocked by translational invariance
THIS WORK
Kernel Decomposition: $\mathbb{R}^{2\times 3} = \Omega \oplus \Omega^\perp$
Ω: translations + rotation; $\Omega^\perp$: inter-peak displacements
$P_\parallel F$: along Ω
$\sum_i \langle w_i, F_i \rangle = 0$
$P_\perp F$: onto $\Omega^\perp$
IFT applicable here
Gauge-Fixing (removes rotational degeneracy):
$\delta y_1 \perp \mathbf{r}_1$ (translation) | $\delta y_3 - \delta y_2 \parallel \mathbf{r}_1$ (rotation)
Open: Verify equivariance + invertibility
IFT on gauge-fixed subspace:
Requires equivariance $F_i(R_\omega \delta y, R) = R_\omega F_i(\delta y, R)$ + invertibility
Lift to PDE via K-N Lemma 2: h_R recovered uniquely
$\Rightarrow$ exact multi-peak ground states ψ_E for all E sufficiently large

2 Background

We wish to study the following nonlinear (time independent) Schrodinger equation (NLS)

$$(-\Delta + V + E)\,\psi + \sigma\,|\psi|^{2p}\,\psi = 0 \tag{2.1}$$

for $V : \mathbb{R}^n \to \mathbb{R}$, $\psi : \mathbb{R}^n \to \mathbb{C}$ with $\sigma \in \mathbb{R}^+$, $E \in \mathbb{R}^+ \cup \{0\}$. For each value of the energy E, we denote solutions of (2.1) as ψ_E. Let $x \in \mathbb{R}^n$, consider

$$u_E\,(x) = E^{-\frac{1}{2p}}\,\psi_E\left(E^{-\frac{1}{2}}x\right), \tag{2.2}$$

from [10] such that

$$\|u_E\|_{H^1}^2 = \|\nabla\psi_E\|_{L^2}^2 + \|\psi_E\|_{L^2}^2$$

remains bounded. By concentration compactness, there exists $M \in \mathbb{N}$ such that

$$\left\|u_E\,(x) - \sum_{i=1}^{M} u_i\,(x - z_i\,(E))\right\|_{H^1} \xrightarrow{E\to\infty} 0, \tag{2.3}$$

where $u_i : \mathbb{R}^n \to \mathbb{C}$ solves

$$(-\Delta + 1)\,u + \sigma\,|u|^{2p}\,u = 0. \tag{2.4}$$

Consider $R = E^{-\frac{1}{2}}$. Under certain decay conditions on the potential V as $|x| \to \infty$, $E^{-\frac{1}{2}}z_i\,(E) = Rz_i\,(R)$ remain bounded and converge to the critical points of V as $R \to 0$.

Claim 1. Let x_0 be a non-degenerate critical point of V, i.e. Hessian at x_0 is invertible.

Claim 2. $Rz_i\,(R) \xrightarrow{R\to 0} x_0$ for $i = 1, \ldots, M$.

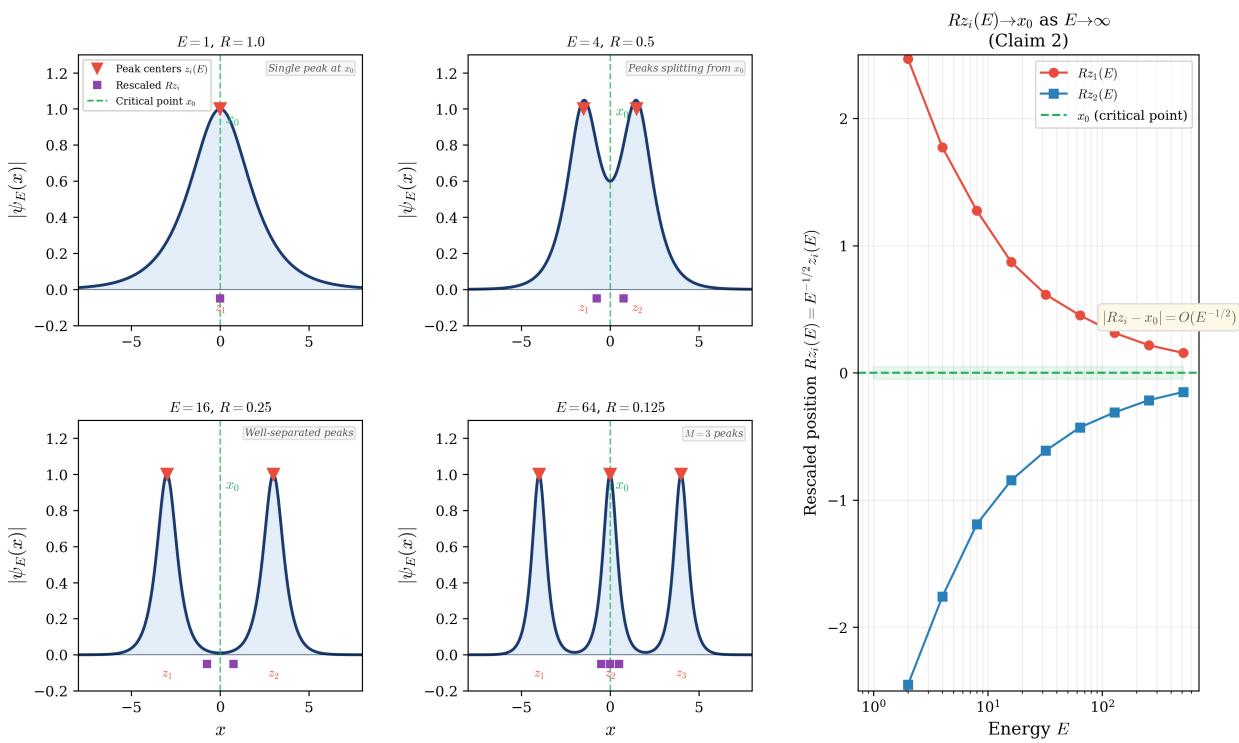


Figure 2.1: graphic

Since $R := E^{-\frac{1}{2}}$, we have

$$\left(-\Delta + R^2 V(Rx) + 1\right) u_E + \sigma |u_E|^{2p} u_E = 0 \tag{2.5}$$

Rewrite (2.3) using the definition of R such that

$$u_E(x) = \sum_{i=1}^{M} u_R(x - z_i(R)) + h_R(x)$$

where now u_R is the only (up to bounded) positive and spherically symmetric solution of

$$\left(-\Delta + R^2 V(Rx_0) + 1\right) u + \sigma |u|^{2p} u = 0.$$

Therefore, $\nabla u_R = \left(\frac{\partial u_R}{\partial x_1}, \ldots, \frac{\partial u_R}{\partial x_n}\right)$ satisfy

$$\left(-\Delta + R^2 V(x_0) + 1\right) \frac{\partial u_R}{\partial x_j} + \sigma (2p+1) |u_R|^{2p} \frac{\partial u_R}{\partial x_j} = 0,$$

for each $j = 1, \ldots, n$.

Lemma 1. *There exists $\varepsilon > 0$ such that if $0 < R < \varepsilon$, then there exists $\tilde{z}_1, \ldots, \tilde{z}_m \in \mathbb{R}^n$ for each R such that $u_E(x) = \underbrace{\sum_{i=1}^{M} u_R(x - z_i(R) - \tilde{z}_i(R))}_{\tilde{u}_E} + \tilde{h}_R(x)$ and $\tilde{h}_R \perp \frac{\partial u_R}{\partial x_j}\left(\cdot - \underbrace{z_i(R) - \tilde{z}_j(R)}_{Now\ z_i(R)}\right)$ for all $i = 1, \ldots, M$ and $j = 1, \ldots, n$.*

Lemma 2. *There exists $\tilde{\varepsilon}, \delta, d > 0$ such that if $R < \tilde{\varepsilon}$, $\|h\|_{H^1} < \delta$ and $\|z_i - z_j\| > d$ for $i, j = 1, \ldots, M$, then*

$$P_\perp F(h, z_1, \ldots, z_M, R) = 0$$

has a unique solution $h = h(z_1, \ldots, z_M, R)$ for all z_i, R where $F = \left(-\Delta + R^2 V(Rx) + 1\right) \Phi_R + \sigma |\Phi_R|^{2p} \Phi_R$ (LHS of (2.5)) and $P_\perp$ is the orthogonal projection.

Consider

$$\left\langle \frac{\partial u_R}{\partial x_j}(x - z_i), F(h(z_1, \ldots, z_M, R), z_1, \ldots, z_M, R) \right\rangle = 0 \tag{2.6}$$

Let $s_i = Rz_i - x_0$ such that (2.6) becomes

$$\|u_R\|_{L^2}^2 \left[H_V(x_0) + O(|s_i|)\right] \frac{s_i}{Rm_R} + \frac{O(1)}{m_R} = \underbrace{\frac{2Q_R(m_R)}{R^4 m_R}}_{\chi} \left[\sum_{k \neq i}^{M} \underbrace{\frac{Q_R(|z_k - z_i|)}{Q_R(m_R)}}_{\alpha_{ki}} \cdot \frac{z_k - z_i}{|z_k - z_i|} + O(1)\right]$$

such that $m_R = \min\{|z_j(R) - z_i(R)| : i \neq j; i, j = 1, \ldots, M\}$, $Q_R(z) = -\sigma \left\langle \frac{\partial u_R}{\partial z/|z|}(x - z), u_R^{2p+1}(x) \right\rangle$, and $\frac{s_i}{Rm_R} = \min\{|s_k - s_j| : i \neq j; i, j = 1, \ldots, M\}$ remain bounded such that $\frac{s_i}{Rm_R} \xrightarrow{R \searrow 0} y_i \in \mathbb{R}^n$,

$$\begin{aligned} H_V(x_0) y_i &= \frac{2Q_R(m_R)}{R^4 m_R} \sum_{k \in N_i} \frac{Q_R(|z_k - z_i|)}{Q_R(m_R)} \frac{z_k - z_i}{|z_k - z_i|} \\ &= \sum_{k \in N_i} \alpha_{ki} \chi \frac{y_k - y_i}{|y_k - y_i|} \\ &= \sum_{k \in N_i} \alpha_{ki} \chi (y_k - y_i) \end{aligned}$$

since $|y_k - y_i| = 1$, and $N_i = \{k = 1, \ldots, M : |y_k - y_i| = 1\}, 0 < \chi, 0 \leq \alpha_{ki} \leq 1$.

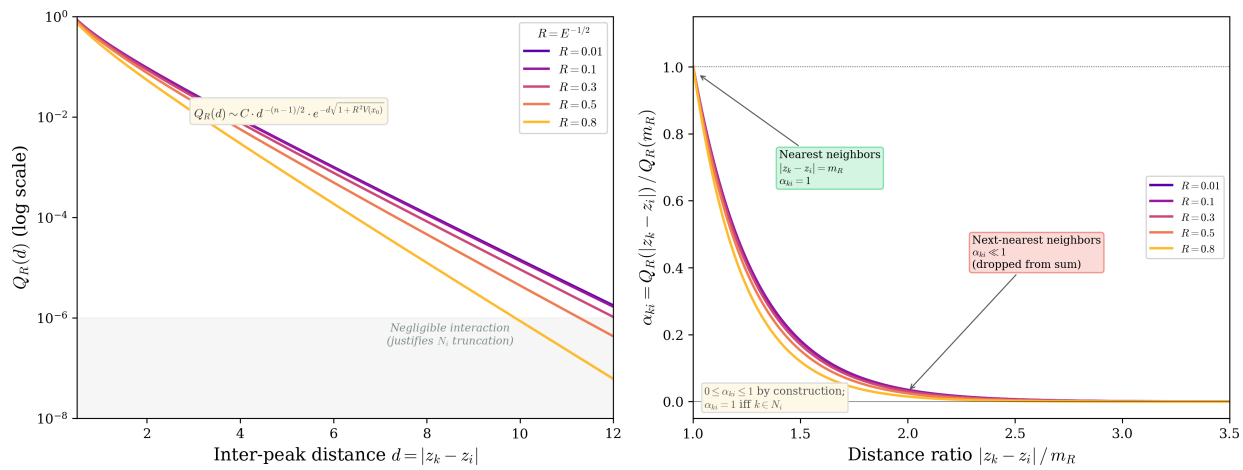


Figure 2.2: (left) Exponential Decay of $Q_R(d) = -\sigma\langle \partial_{\hat{d}} u_R(\cdot - d), u_R^{2p+1}\rangle$
(right) Interaction Coefficient α_{ki} vs Relative Distance.

Since $y_i = \lim_{R\to 0} \frac{s_i}{Rm_R}$, we know:

Corollary 1. *Let $U = span\{\boldsymbol{r} \in \mathbb{R}^n | eigvec(H_V) \ni \boldsymbol{r}, eigval(H_V) < 0\}$ be the so-called "unstable manifold" of H_V, then $span\{y_1, \ldots, y_m\} \subseteq U$ and are invariant under H_V.*

3 Summary of Scenarios

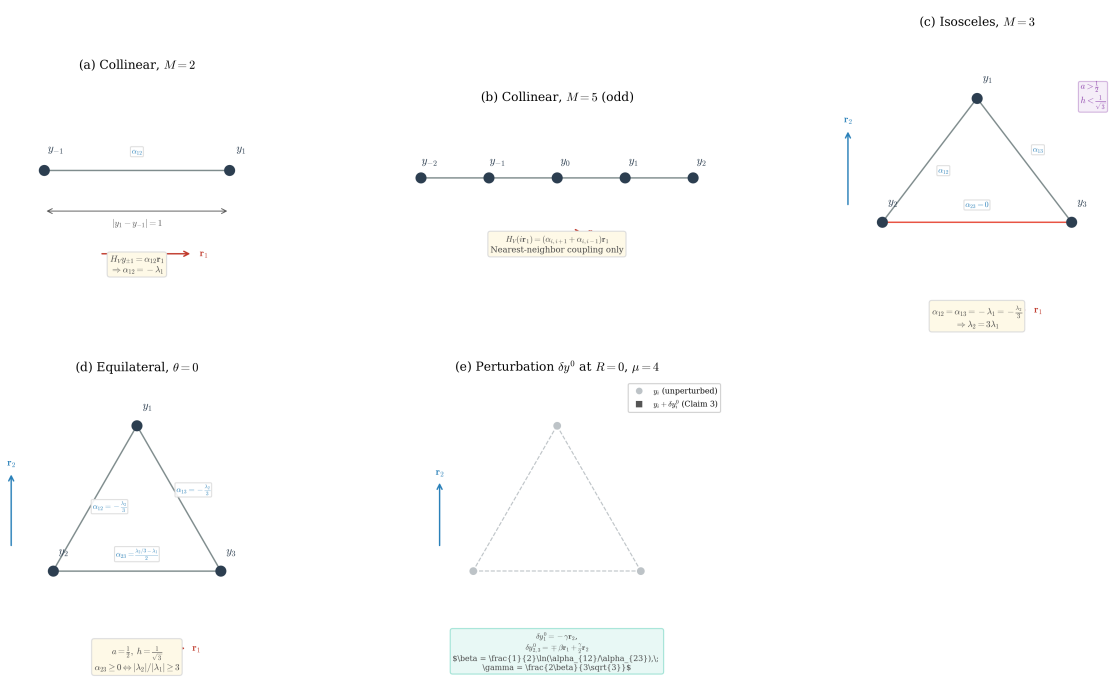


Figure 3.1: Peak Configurations

Example 1. Collinear

Let y_i denote the n connected, collinear, points lying along the vector $\boldsymbol{r_1} \in \mathbb{R}$. For n odd: let $y_0 = 0$ and $y_{\pm i} = \pm i \boldsymbol{r_1}$ such that $H_V y_i = \alpha_{i,i+1}(y_{i+1} - y_i) + \alpha_{i,i-1}(y_i - y_{i-1})$. For n even: let $y_{\pm i} = \pm(i - 1/2)\boldsymbol{r_1}$ such that $y_i = (i - 1/2)\boldsymbol{r_1}$ and $y_{-i} = (-i + 1/2)\boldsymbol{r_1}$; similarly, $H_V y_i = \alpha_{i,i+1}(y_{i+1} - y_i) + \alpha_{i,i-1}(y_i - y_{i-1})$. Therefore

$$\begin{cases} H_V (i - 1/2)\boldsymbol{r_1} = (\alpha_{i,i+1} + \alpha_{i,i-1})\boldsymbol{r_1}\, \forall i \leq n \in \mathbb{N}, & n \text{ even} \\ H_V i \boldsymbol{r_1} = (\alpha_{i,i+1} + \alpha_{i,i-1})\boldsymbol{r_1}\, \forall i \leq n \in \mathbb{N}, & n \text{ odd} \end{cases}$$

Example 2. Coplanar ($n = 3$, Isosceles)

Let y_1, y_2, y_3 denote the connected, coplanar, points lying in the plane spanned by vectors $\boldsymbol{r_1}, \boldsymbol{r_2} \in \mathbb{R}^2$; $H_V \boldsymbol{r_i} = \lambda_i \boldsymbol{r_i}, \lambda_i < 0, |\boldsymbol{r_i}| = 1\, \forall i = 1, 2$. Rotate the triangle such that y_1 lies along $\boldsymbol{r_2}$ by convention. We can then write

$$\begin{aligned} y_1 &= (0, h)^\intercal & y_2 &= \left(-a, -\frac{h}{2}\right)^\intercal & y_3 &= \left(a, -\frac{h}{2}\right)^\intercal \\ &= h\boldsymbol{r_2} & &= -a\boldsymbol{r_1} - \left(\frac{h}{2}\right)\boldsymbol{r_2} & &= a\boldsymbol{r_1} - \left(\frac{h}{2}\right)\boldsymbol{r_2} \end{aligned}$$

such that

$$\begin{cases} 2a > 1 \\ a^2 + \left(\frac{3h}{2}\right)^2 = 1 \end{cases} \implies \begin{cases} a > \frac{1}{2} \\ \left(\frac{3h}{2}\right)^2 = 1 - a^2 < \frac{3}{4} \end{cases} \implies \begin{cases} a > \frac{1}{2} \\ h < \frac{1}{\sqrt{3}} \end{cases}$$

Now

$$\begin{aligned} H_V y_1 &= \alpha_{12}(y_2 - y_1) + \alpha_{13}(y_3 - y_1) \\ (0, \lambda_2 h)^\intercal &= \alpha_{12}\left(-a, -\frac{3h}{2}\right)^\intercal + \alpha_{13}\left(a, -\frac{3h}{2}\right)^\intercal \\ &= \left(a(\alpha_{13} - \alpha_{12}), -\frac{3h}{2}(\alpha_{13} + \alpha_{12})\right)^\intercal \end{aligned}$$

$$\Downarrow$$

$$\begin{cases} a(\alpha_{13} - \alpha_{12}) = & 0 \\ -\frac{3h}{2}(\alpha_{13} + \alpha_{12}) = & \lambda_2 h \end{cases} \implies \alpha_{12} = \alpha_{13} = -\frac{\lambda_2}{3}$$

Similarly

$$\begin{aligned} H_V y_2 &= \alpha_{12}(y_1 - y_2) \\ \left(-a\lambda_1, -\left(\frac{h}{2}\right)\lambda_2\right)^\intercal &= \alpha_{12}\left(a, \frac{3h}{2}\right)^\intercal \end{aligned}$$

$$\Downarrow$$

$$\begin{cases} a\alpha_{12} = & -a\lambda_1 \\ \frac{3h}{2}\alpha_{12} = & -\left(\frac{h}{2}\right)\lambda_2 \end{cases} \implies \alpha_{12} = -\lambda_1 = -\frac{\lambda_2}{3} \implies \lambda_2 = 3\lambda_1$$

(the case for y_3 is superfluous).

Example 3. Coplanar ($n = 3$, Equilateral)

Let y_1, y_2, y_3 denote the connected, coplanar, points lying in the plane spanned by vectors $\boldsymbol{r_1}, \boldsymbol{r_2} \in \mathbb{R}^2$; $H_V \boldsymbol{r_i} = \lambda_i \boldsymbol{r_i}, \lambda_i < 0, |\boldsymbol{r_i}| = 1\, \forall i = 1, 2$. Rotate the triangle such that y_1 lies along $\boldsymbol{r_2}$ by convention. We can then write

$$\begin{aligned} y_1 &= (0, h)^\intercal & y_2 &= \left(-a, -\frac{h}{2}\right)^\intercal & y_3 &= \left(a, -\frac{h}{2}\right)^\intercal \\ &= h\boldsymbol{r_2} & &= -a\boldsymbol{r_1} - \left(\frac{h}{2}\right)\boldsymbol{r_2} & &= a\boldsymbol{r_1} - \left(\frac{h}{2}\right)\boldsymbol{r_2} \end{aligned}$$

such that

$$\left\{\begin{array}{c} 2a = 1 \\ a^2 + \left(\frac{3h}{2}\right)^2 = 1 \end{array}\right. \implies \left\{\begin{array}{c} a = \frac{1}{2} \\ \left(\frac{3h}{2}\right)^2 = 1 - a^2 \end{array}\right. \implies \left\{\begin{array}{c} a = \frac{1}{2} \\ h = \frac{1}{\sqrt{3}} \end{array}\right.$$

Now

$$\begin{aligned} H_V y_1 &= \alpha_{12}\left(y_2 - y_1\right) + \alpha_{13}\left(y_3 - y_1\right) \\ \left(0, \lambda_2 h\right)^\intercal &= \alpha_{12}\left(-a, -\frac{3h}{2}\right)^\intercal + \alpha_{13}\left(a, -\frac{3h}{2}\right)^\intercal \\ &= \left(a\left(\alpha_{13} - \alpha_{12}\right), -\frac{3h}{2}\left(\alpha_{13} + \alpha_{12}\right)\right)^\intercal \end{aligned}$$

$$\Downarrow$$

$$\begin{cases} a\left(\alpha_{13} - \alpha_{12}\right) = & 0 \\ -\frac{3h}{2}\left(\alpha_{13} + \alpha_{12}\right) = & \lambda_2 h \end{cases} \implies \alpha_{12} = \alpha_{13} = -\frac{\lambda_2}{3}$$

Similarly

$$\begin{aligned} H_V y_2 &= \alpha_{12}\left(y_1 - y_2\right) + \alpha_{23}\left(y_3 - y_2\right) \\ \left(-a\lambda_1, -\left(\frac{h}{2}\right)\lambda_2\right)^\intercal &= \alpha_{12}\left(a, \frac{3h}{2}\right)^\intercal + \alpha_{23}\left(2a, 0\right)^\intercal \\ &= \left(a\left(\alpha_{12} + 2\alpha_{23}\right), \left(\frac{3h}{2}\right)\alpha_{12}\right)^\intercal \end{aligned}$$

$$\Downarrow$$

$$\begin{cases} a\left(\alpha_{12} + 2\alpha_{23}\right) = & -a\lambda_1 \\ \left(\frac{3h}{2}\right)\alpha_{12} = & -\left(\frac{h}{2}\right)\lambda_2 \end{cases} \implies \begin{cases} \alpha_{12} = & -\frac{\lambda_2}{3} \\ \alpha_{23} = & \frac{\lambda_2}{6} - \frac{\lambda_1}{2} \end{cases}$$

(the case for y_3 is superfluous). Since we require each $\alpha \geq 0$ we can readily see

$$\begin{aligned} \alpha_{23} &\geq 0 \\ \frac{\lambda_2}{6} - \frac{\lambda_1}{2} &\geq \\ \lambda_2 &\geq 3\lambda_1 \\ \left|\lambda_2\right| &\geq 3\left|\lambda_1\right| \\ \frac{\left|\lambda_2\right|}{\left|\lambda_1\right|} &\geq 3 \end{aligned}$$

Example 4. Coplanar ($n = 3$, Equilateral, Rotational DOF θ)

Let $R\left(\theta\right) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ be the 2-dimensional rotation matrix with inverse $R^{-1}\left(\theta\right) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$. Let y_1, y_2, y_3 denote the connected, coplanar, points lying in the plane spanned by vectors $\boldsymbol{r_1}, \boldsymbol{r_2} \in \mathbb{R}^2$; $H_V \boldsymbol{r_i} = \lambda_i \boldsymbol{r_i}, \lambda_i < 0, \left|\boldsymbol{r_i}\right| = 1\, \forall i = 1, 2$. Rotate the triangle such that y_1 lies along $\boldsymbol{r_2}$ by convention. We can then write

$$\begin{aligned} y_1' &= \left(0, h\right)^\intercal & y_2' &= \left(-a, -\frac{h}{2}\right)^\intercal & y_3' &= \left(a, -\frac{h}{2}\right)^\intercal \\ &= h\boldsymbol{r_2} & &= -a\boldsymbol{r_1} - \left(\frac{h}{2}\right)\boldsymbol{r_2} & &= a\boldsymbol{r_1} - \left(\frac{h}{2}\right)\boldsymbol{r_2} \end{aligned}$$

such that

$$\left\{\begin{array}{c} 2a = 1 \\ a^2 + \left(\frac{3h}{2}\right)^2 = 1 \end{array}\right. \implies \left\{\begin{array}{c} a = \frac{1}{2} \\ \left(\frac{3h}{2}\right)^2 = 1 - a^2 \end{array}\right. \implies \left\{\begin{array}{c} a = \frac{1}{2} \\ h = \frac{1}{\sqrt{3}} \end{array}\right.$$

as well as

$$
\begin{aligned}
y_1 = R(\theta)\, y_1' &= \begin{pmatrix} -h\sin\theta \\ h\cos\theta \end{pmatrix} = -h\sin(\theta)\,\boldsymbol{r_1} + h\cos(\theta)\,\boldsymbol{r_2} \\
y_2 = R(\theta)\, y_2' &= \begin{pmatrix} -a\cos\theta + \left(\frac{h}{2}\right)\sin\theta \\ -a\sin\theta - \left(\frac{h}{2}\right)\cos\theta \end{pmatrix} = \left(-a\cos\theta + \left(\frac{h}{2}\right)\sin\theta\right)\boldsymbol{r_1} - \left(a\sin\theta + \left(\frac{h}{2}\right)\cos\theta\right)\boldsymbol{r_2} \\
y_3 = R(\theta)\, y_3' &= \begin{pmatrix} a\cos\theta + \left(\frac{h}{2}\right)\sin\theta \\ a\sin\theta - \left(\frac{h}{2}\right)\cos\theta \end{pmatrix} = \left(a\cos\theta + \left(\frac{h}{2}\right)\sin\theta\right)\boldsymbol{r_1} + \left(a\sin\theta - \left(\frac{h}{2}\right)\cos\theta\right)\boldsymbol{r_2}
\end{aligned}
$$

Now

$$
\begin{aligned}
H_V y_1 &= \alpha_{12}(y_2 - y_1) + \alpha_{13}(y_3 - y_1) \\
H_V R(\theta)\, y_1' &= \\
R^{-1}(\theta)\, H_V R(\theta)\, y_1' &= \alpha_{12}(y_2' - y_1') + \alpha_{13}(y_3' - y_1') \\
&\Downarrow \\
\alpha_{12}\begin{pmatrix} -a \\ -\frac{3h}{2} \end{pmatrix} + \alpha_{13}\begin{pmatrix} a \\ -\frac{3h}{2} \end{pmatrix} &= \begin{pmatrix} \lambda_2\sin^2\theta + \lambda_1\cos^2\theta & (\lambda_2 - \lambda_1)\sin\theta\cos\theta \\ (\lambda_2 - \lambda_1)\sin\theta\cos\theta & \lambda_2\cos^2\theta + \lambda_1\sin^2\theta \end{pmatrix}\begin{pmatrix} 0 \\ h \end{pmatrix} \\
\begin{pmatrix} \alpha_{13} - \alpha_{12} \\ \alpha_{13} + \alpha_{12} \end{pmatrix} &= \begin{pmatrix} \frac{h}{a}(\lambda_2 - \lambda_1)\sin\theta\cos\theta \\ -\frac{2}{3}\left(\lambda_2\cos^2\theta + \lambda_1\sin^2\theta\right) \end{pmatrix} \\
\alpha_{12} &= \frac{1}{3}\left(\lambda_2\left(-\sqrt{3}\sin\theta - \cos\theta\right)\cos\theta + \lambda_1\left(-\sin\theta + \sqrt{3}\cos\theta\right)\sin\theta\right)
\end{aligned}
$$

$$
\begin{aligned}
H_V y_2 &= \alpha_{12}(y_1 - y_2) + \alpha_{23}(y_3 - y_2) \\
H_V R(\theta)\, y_2' &= \\
R^{-1}(\theta)\, H_V R(\theta)\, y_2' &= \alpha_{12}(y_1' - y_2') + \alpha_{23}(y_3' - y_2') \\
&\Downarrow \\
\alpha_{12}\begin{pmatrix} a \\ \frac{3h}{2} \end{pmatrix} + \alpha_{23}\begin{pmatrix} 2a \\ 0 \end{pmatrix} &= \begin{pmatrix} \lambda_2\sin^2\theta + \lambda_1\cos^2\theta & (\lambda_2 - \lambda_1)\sin\theta\cos\theta \\ (\lambda_2 - \lambda_1)\sin\theta\cos\theta & \lambda_2\cos^2\theta + \lambda_1\sin^2\theta \end{pmatrix}\begin{pmatrix} -a \\ -\frac{h}{2} \end{pmatrix} \\
\begin{pmatrix} \alpha_{12} + 2\alpha_{23} \\ 3\alpha_{12} \end{pmatrix} &= \begin{pmatrix} -\left(\lambda_2\sin^2\theta + \lambda_1\cos^2\theta\right) - \frac{h}{2a}\left((\lambda_2 - \lambda_1)\sin\theta\cos\theta\right) \\ -\left(\lambda_2\cos^2\theta + \lambda_1\sin^2\theta\right) - \frac{2a}{h}(\lambda_2 - \lambda_1)\sin\theta\cos\theta \end{pmatrix} \\
\alpha_{13} &= \frac{1}{3}\left(\lambda_2\left(\sqrt{3}\sin\theta - \cos\theta\right)\cos\theta - \lambda_1\left(\sin\theta + \sqrt{3}\cos\theta\right)\sin\theta\right)
\end{aligned}
$$

$$
\begin{aligned}
H_V y_3 &= \alpha_{13}(y_1 - y_3) + \alpha_{23}(y_2 - y_3) \\
H_V R(\theta)\, y_3' &= \\
R^{-1}(\theta)\, H_V R(\theta)\, y_3' &= \alpha_{13}(y_1' - y_3') + \alpha_{23}(y_2' - y_3') \\
&\Downarrow \\
\alpha_{13}\begin{pmatrix} -a \\ \frac{3h}{2} \end{pmatrix} + \alpha_{23}\begin{pmatrix} -2a \\ 0 \end{pmatrix} &= \begin{pmatrix} \lambda_2\sin^2\theta + \lambda_1\cos^2\theta & (\lambda_2 - \lambda_1)\sin\theta\cos\theta \\ (\lambda_2 - \lambda_1)\sin\theta\cos\theta & \lambda_2\cos^2\theta + \lambda_1\sin^2\theta \end{pmatrix}\begin{pmatrix} a \\ -\frac{h}{2} \end{pmatrix} \\
\begin{pmatrix} \alpha_{13} + 2\alpha_{23} \\ 3\alpha_{13} \end{pmatrix} &= \begin{pmatrix} -\left(\lambda_2\sin^2\theta + \lambda_1\cos^2\theta\right) + \frac{h}{2a}(\lambda_2 - \lambda_1)\sin\theta\cos\theta \\ -\left(\lambda_2\cos^2\theta + \lambda_1\sin^2\theta\right) + \frac{2a}{h}(\lambda_2 - \lambda_1)\sin\theta\cos\theta \end{pmatrix} \\
\alpha_{23} &= \frac{1}{6}\left(\lambda_2\left(2\cos(2\theta) - 1\right) - \lambda_1\left(2\cos(2\theta) + 1\right)\right)
\end{aligned}
$$

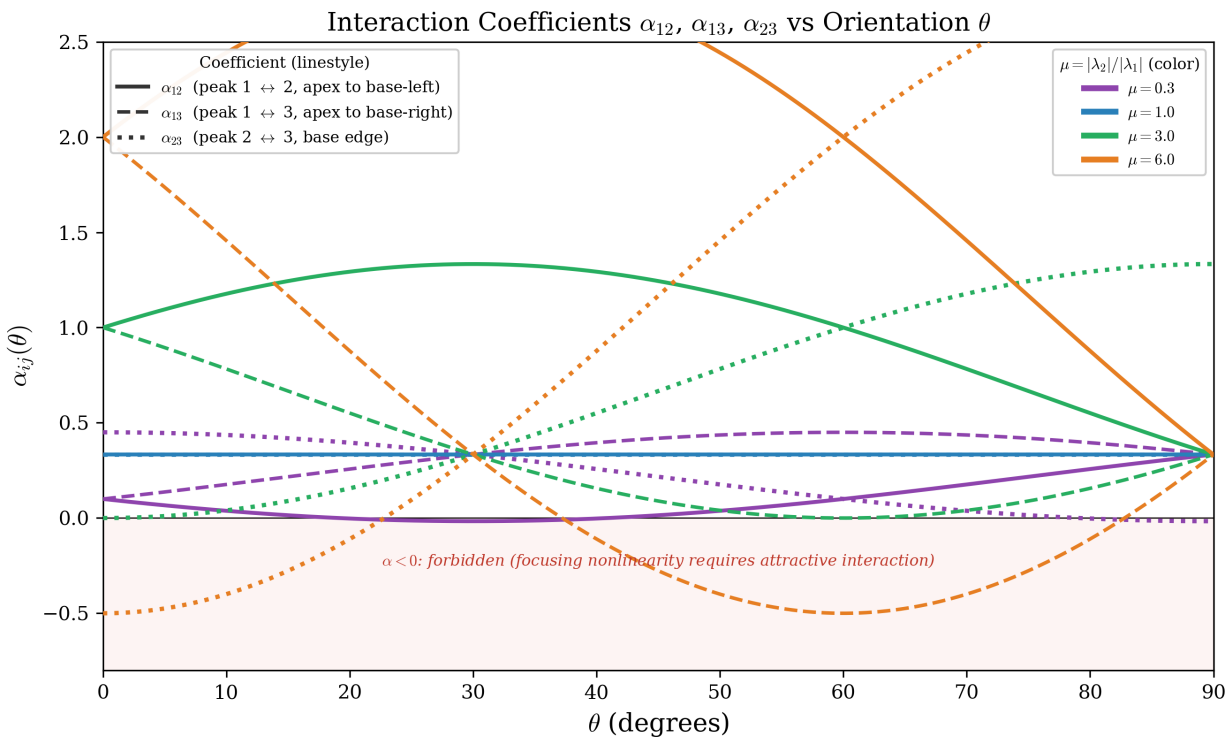


Figure 3.2: α vs. θ plots for various μ

We again assert each $\alpha \geq 0$ which constrains θ for different ranges of $\frac{|\lambda_2|}{|\lambda_1|}$. Define $\mu = \frac{|\lambda_2|}{|\lambda_1|}$, note that $\lambda_i = \operatorname{sign}(\lambda_i)\,|\lambda_i|\ \forall i \in \{1,2\}$ by definition, and that $\lambda_i \leq 0\,\forall i \in \{1,2\} \Rightarrow 0 < \mu < \infty; \alpha_{12}, \alpha_{13}, \alpha_{23} \geq 0; 0 \leq \theta \leq \frac{\pi}{2}$ by assumption. We therefore seek admissible solutions of

$$\begin{cases} \frac{1}{3}\left(\lambda_2\left(-\sqrt{3}\sin\theta - \cos\theta\right)\cos\theta + \lambda_1\left(-\sin\theta + \sqrt{3}\cos\theta\right)\sin\theta\right) & \geq 0 \\ \frac{1}{3}\left(\lambda_2\left(+\sqrt{3}\sin\theta - \cos\theta\right)\cos\theta - \lambda_1\left(+\sin\theta + \sqrt{3}\cos\theta\right)\sin\theta\right) & \geq 0 \\ \frac{1}{6}\left(\lambda_2\left(2\cos(2\theta) - 1\right) - \lambda_1\left(2\cos(2\theta) + 1\right)\right) & \geq 0 \end{cases}$$

$$\Downarrow$$

$$\begin{cases} \sin^2\theta + \mu\cos^2\theta + \sqrt{3}\,(\mu - 1)\sin\theta\cos\theta & \geq 0 \quad \text{(I)} \\ \sin^2\theta + \mu\cos^2\theta - \sqrt{3}\,(\mu - 1)\sin\theta\cos\theta & \geq 0 \quad \text{(II)} \\ (\mu + 1) - 2\,(\mu - 1)\cos(2\theta) & \geq 0 \quad \text{(III)} \end{cases}$$

Since $\cos^2\theta = \frac{1}{2}\left(1 + \cos(2\theta)\right)$, $\sin^2\theta = \frac{1}{2}\left(1 - \cos(2\theta)\right)$, and $\sin\theta\cos\theta = \frac{1}{2}\sin(2\theta)$, (I) & (II) are equivalent to

$$(\mu + 1) + (\mu - 1)\cos(2\theta) \pm \sqrt{3}\,(\mu - 1)\sin(2\theta) \geq 0$$

$$\Updownarrow$$

$$(\mu + 1) + 2\,(\mu - 1)\cos\left(2\theta \mp \frac{\pi}{3}\right) \geq 0 \quad \text{(IV)}, \text{(V)}$$

(III), (IV), and (V) require

$$(\mu + 1) + 2\,(\mu - 1)\cos\phi \geq 0 \quad \phi \in \left\{2\theta - \frac{\pi}{3}, 2\theta, 2\theta + \frac{\pi}{3}\right\}$$

Let $\nu(\mu) = \frac{\mu+1}{2|\mu-1|}$ and $\varphi(\mu) = \arccos \nu(\mu)$ such that

$$\varphi(\mu) = \begin{cases} \frac{\pi}{3}, & \nu(\mu) \leq \frac{1}{2} \\ 0 < \varphi(\mu) < \frac{\pi}{3}, & \frac{1}{2} < \nu(\mu) < 1 \\ 0, & 1 \leq \nu(\mu) \end{cases}$$

For $0 < \mu < \frac{1}{3}$ we know $\mu - 1 < 0 \Rightarrow \nu(\mu) = \frac{\mu+1}{2(1-\mu)} \in \left(\frac{1}{2}, 1\right)$. We have

$$\varphi(\mu) = \arccos\left(\frac{\mu+1}{2(1-\mu)}\right) \in \left(0, \frac{\pi}{3}\right)$$

such that

$$\theta \in \left[0, \frac{\frac{\pi}{3} - \varphi(\mu)}{2}\right] \cup \left[\frac{\frac{\pi}{3} + \varphi(\mu)}{2}, \frac{\pi - \varphi(\mu)}{2}\right]$$

For $\frac{1}{3} \leq \mu \leq 3$ we have all θ admissible, i.e.

$$\theta \in \left[0, \frac{\pi}{2}\right]$$

For $3 < \mu$ we know $\mu - 1 > 0 \Rightarrow \nu(\mu) = \frac{\mu+1}{2(\mu-1)} \in \left(\frac{1}{2}, 1\right)$. We have

$$\varphi(\mu) = \arccos\left(\frac{\mu+1}{2(\mu-1)}\right) \in \left(0, \frac{\pi}{3}\right)$$

such that

$$\theta \in \left[\frac{\varphi(\mu)}{2}, \frac{\frac{2\pi}{3} - \varphi(\mu)}{2}\right] \cup \left[\frac{\frac{2\pi}{3} + \varphi(\mu)}{2}, \frac{\pi}{2}\right]$$

Table 3.1: Summary of Results

$\mu-$Range	$\theta-$Range
$0 < \mu < \frac{1}{3}$	$\left[0, \frac{\pi}{6} - \frac{\varphi(\mu)}{2}\right] \cup \left[\frac{\pi}{6} + \frac{\varphi(\mu)}{2}, \frac{\pi}{2} - \frac{\varphi(\mu)}{2}\right]$
$\frac{1}{3} \leq \mu \leq 3$	$\left[0, \frac{\pi}{2}\right]$
$3 < \mu$	$\left[\frac{\varphi(\mu)}{2}, \frac{\pi}{3} - \frac{\varphi(\mu)}{2}\right] \cup \left[\frac{\pi}{3} + \frac{\varphi(\mu)}{2}, \frac{\pi}{2}\right]$

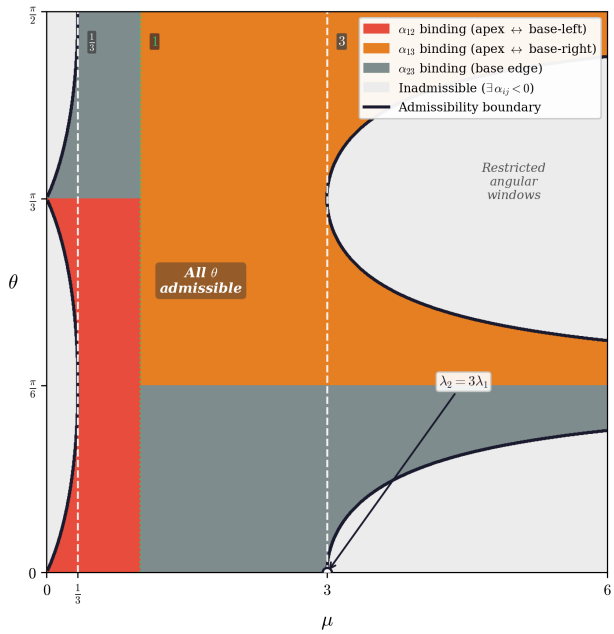


Figure 3.3: Admissible Configurations in (μ, θ) Space: Binding Constraint Identification

4 Research Problem

We seek solutions of

$$H_V \frac{\tilde{y}_i}{\tilde{m}_R} = \sum_{k \in N_i} \frac{2Q_R\left(|\tilde{y}_k - \tilde{y}_i|\right)}{\|u_R\|_{L^2}^2 R^4 |\tilde{y}_k - \tilde{y}_i|} \cdot \frac{\tilde{y}_k - \tilde{y}_i}{\tilde{m}_R} \tag{4.1}$$

such that $\frac{\tilde{y}_i}{\tilde{m}_R} \xrightarrow[R \searrow 0]{} y_i$ and

$$\frac{2Q_R\left(|\tilde{y}_k - \tilde{y}_i|\right)}{\|u_R\|_{L^2}^2 R^4 |\tilde{y}_k - \tilde{y}_i|} \xrightarrow[R \searrow 0]{} \tilde{\alpha}_{ki}.$$

We write $\tilde{y}_i = \tilde{m}_R y_i + \delta y_i$ (such that $\frac{\delta y_i}{\tilde{m}_R} \xrightarrow[R \searrow 0]{} 0$), $\delta y_i \in \operatorname{span}\{\boldsymbol{r_1}, \boldsymbol{r_2}\} \forall i = \overline{1,3}$, $R > 0$. Define

$$F_i\left(\delta y, R\right) = H_V y_i + H_V \frac{\delta y_i}{\tilde{m}_R} - \sum_{k \in N_i} \frac{2Q_R\left(|\tilde{m}_R\left(y_k - y_i\right) + \delta y_k - \delta y_i|\right)}{\|u_R\|_{L^2}^2 R^4 |\tilde{m}_R\left(y_k - y_i\right) + \delta y_k - \delta y_i|} \left(y_k - y_i + \frac{\delta y_k - \delta y_i}{\tilde{m}_R}\right), \tag{4.2}$$

and extend F_i at $R = 0$ by continuity

$$F_i\left(\delta y, 0\right) = H_V y_i - \sum_{k \in N_i} \underbrace{e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle} \chi}_{\alpha_{ki} = \tilde{\alpha}_{ki} \chi} \left(y_k - y_i\right), \tag{4.3}$$

with derivative along a vector $w \in \operatorname{span}\{\boldsymbol{r_1}, \boldsymbol{r_2}\}$

$$D_{\delta y} F_i\left(\delta y, 0\right)[w] = \sum_{k \in N_i} \langle w_k - w_i, y_k - y_i \rangle e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle} \chi\left(y_k - y_i\right). \tag{4.4}$$

With the ultimate goal of utilizing the implicit function theorem (IFT) we must start with a solution of

$$F_i\left(\delta y, 0\right) = 0 \quad \forall i = \overline{1,3} \tag{4.5}$$

Claim 3. $\delta y^0 = \left(\delta y_1^0, \delta y_2^0, \delta y_3^0\right)$, $\delta y_1^0 = -\gamma \boldsymbol{r_2}$, $\delta y_{2,3}^0 = \mp \beta \boldsymbol{r_1} + \frac{\gamma}{2} \boldsymbol{r_2}$, with $\beta = \frac{1}{2} \ln\left(\frac{\alpha_{12}}{\alpha_{23}}\right)$, $\gamma = \frac{2\beta}{3\sqrt{3}}$ solves (4.5).

Proof. Plugging the given ansatz into (4.5) we obtain

$$\begin{cases} F_1 = & H_V y_1 - \left(e^{-\langle y_2-y_1,\delta y_2-\delta y_1\rangle}\left(y_2-y_1\right) + e^{-\langle y_3-y_1,\delta y_3-\delta y_1\rangle}\left(y_3-y_1\right)\right)\chi \\ F_2 = & H_V y_2 - \left(e^{-\langle y_1-y_2,\delta y_1-\delta y_2\rangle}\left(y_1-y_2\right) + e^{-\langle y_3-y_2,\delta y_3-\delta y_2\rangle}\left(y_3-y_2\right)\right)\chi \\ F_3 = & H_V y_3 - \left(e^{-\langle y_1-y_3,\delta y_1-\delta y_3\rangle}\left(y_1-y_3\right) + e^{-\langle y_2-y_3,\delta y_2-\delta y_3\rangle}\left(y_2-y_3\right)\right)\chi \end{cases}$$

With $\tilde{\alpha}_{ki} = e^{-\langle y_k-y_i,\delta y_k-\delta y_i\rangle} \Rightarrow \tilde{\alpha}_{ki} = \tilde{\alpha}_{ik}$, we find $\tilde{\alpha}_{12} = \tilde{\alpha}_{21} = \tilde{\alpha}_{13} = \tilde{\alpha}_{31} = 1$ and $\tilde{\alpha}_{23} = \tilde{\alpha}_{32} = \frac{\alpha_{23}}{\alpha_{12}}$. For the equilateral triangle we know $\alpha_{12} = \alpha_{13} = -\frac{\lambda_2}{3} > \alpha_{23} = \frac{-\lambda_1+\frac{\lambda_2}{3}}{2} \Rightarrow \chi = \max\{\alpha_{12},\alpha_{13},\alpha_{23}\} = \alpha_{12} = \alpha_{13}$ and $H_V = \begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix}, y_1 = \left(0, \frac{1}{\sqrt{3}}\right)^\intercal, y_2 = \left(-\frac{1}{2}, -\frac{1}{2\sqrt{3}}\right)^\intercal, y_3 = \left(\frac{1}{2}, -\frac{1}{2\sqrt{3}}\right)^\intercal$. Therefore

$$\begin{cases} H_V y_1 - \left(\tilde{\alpha}_{21}\left(y_2-y_1\right) + \tilde{\alpha}_{31}\left(y_3-y_1\right)\right)\chi & = 0 \\ H_V y_2 - \left(\tilde{\alpha}_{12}\left(y_1-y_2\right) + \tilde{\alpha}_{32}\left(y_3-y_2\right)\right)\chi & = 0 \\ H_V y_3 - \left(\tilde{\alpha}_{13}\left(y_1-y_3\right) + \tilde{\alpha}_{23}\left(y_2-y_3\right)\right)\chi & = 0 \end{cases} \Rightarrow \begin{cases} H_V y_1 - \left(y_2-y_1+y_3-y_1\right)\chi & = 0 \\ H_V y_2 - \left(y_1-y_2+\frac{\alpha_{23}}{\alpha_{12}}\left(y_3-y_2\right)\right)\chi & = 0 \\ H_V y_3 - \left(y_1-y_3+\frac{\alpha_{23}}{\alpha_{12}}\left(y_2-y_3\right)\right)\chi & = 0 \end{cases}$$

$$\Downarrow$$

$$\begin{cases} \begin{pmatrix}0\\ \frac{\lambda_2}{\sqrt{3}}\end{pmatrix} - \left[\begin{pmatrix}-\frac{1}{2}\\ -\frac{\sqrt{3}}{2}\end{pmatrix} + \begin{pmatrix}\frac{1}{2}\\ -\frac{\sqrt{3}}{2}\end{pmatrix}\right]\chi & = \mathbf{0} \\ \begin{pmatrix}-\frac{\lambda_1}{2}\\ -\frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} - \left[\begin{pmatrix}\frac{1}{2}\\ \frac{\sqrt{3}}{2}\end{pmatrix} + \frac{\alpha_{23}}{\alpha_{12}}\begin{pmatrix}1\\ 0\end{pmatrix}\right]\chi & = \mathbf{0} \\ \begin{pmatrix}\frac{\lambda_1}{2}\\ -\frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} - \left[\begin{pmatrix}-\frac{1}{2}\\ \frac{\sqrt{3}}{2}\end{pmatrix} + \frac{\alpha_{23}}{\alpha_{12}}\begin{pmatrix}-1\\ 0\end{pmatrix}\right]\chi & = \mathbf{0} \end{cases} \Rightarrow \begin{cases} \begin{pmatrix}0\\ \frac{\lambda_2}{\sqrt{3}}\end{pmatrix} + \begin{pmatrix}0\\ \sqrt{3}\alpha_{12}\end{pmatrix} & = \mathbf{0} \\ \begin{pmatrix}-\frac{\lambda_1}{2}\\ -\frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} - \begin{pmatrix}\frac{\alpha_{12}}{2}\\ \frac{\sqrt{3}\alpha_{12}}{2}\end{pmatrix} - \begin{pmatrix}\alpha_{23}\\ 0\end{pmatrix} & = \mathbf{0} \\ \begin{pmatrix}\frac{\lambda_1}{2}\\ -\frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} - \begin{pmatrix}-\frac{\alpha_{12}}{2}\\ \frac{\sqrt{3}\alpha_{12}}{2}\end{pmatrix} + \begin{pmatrix}\alpha_{23}\\ 0\end{pmatrix} & = \mathbf{0} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} \begin{pmatrix}0\\ \frac{\lambda_2}{\sqrt{3}}\end{pmatrix} + \begin{pmatrix}0\\ -\frac{\sqrt{3}\lambda_2}{3}\end{pmatrix} & = \mathbf{0} \\ \begin{pmatrix}-\frac{\lambda_1}{2}\\ -\frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} - \begin{pmatrix}-\frac{\lambda_2}{6}\\ -\frac{\sqrt{3}\lambda_2}{6}\end{pmatrix} - \begin{pmatrix}\frac{-\lambda_1+\frac{\lambda_2}{3}}{2}\\ 0\end{pmatrix} & = \mathbf{0} \\ \begin{pmatrix}\frac{\lambda_1}{2}\\ -\frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} - \begin{pmatrix}\frac{\lambda_2}{6}\\ -\frac{\sqrt{3}\lambda_2}{6}\end{pmatrix} + \begin{pmatrix}\frac{-\lambda_1+\frac{\lambda_2}{3}}{2}\\ 0\end{pmatrix} & = \mathbf{0} \end{cases} \Rightarrow \begin{cases} \begin{pmatrix}0\\ \frac{\lambda_2}{\sqrt{3}} - \frac{\lambda_2}{\sqrt{3}}\end{pmatrix} & = \mathbf{0} \\ \begin{pmatrix}\frac{\lambda_1}{2} - \frac{\lambda_1}{2}\\ \frac{\lambda_2}{2\sqrt{3}} - \frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} & = \mathbf{0} \ \checkmark \\ \begin{pmatrix}\frac{\lambda_1}{2} - \frac{\lambda_1}{2}\\ \frac{\lambda_2}{2\sqrt{3}} - \frac{\lambda_2}{2\sqrt{3}}\end{pmatrix} & = \mathbf{0} \end{cases}$$

$\therefore \delta y^0$ as stated above solves (4.5) for the equilateral triangle. □

Claim 4. (4.4) has a nontrivial kernel (w_1, w_2, w_3) so the IFT does not apply.

Proof. Let

$$\begin{aligned} \Omega &= \ker\left(D_{\delta y}F_i\left(\delta y^0, 0\right)\right) \\ &= \operatorname{span}\left\{(w_1,w_2,w_3) \mid \langle w_k - w_i, y_k - y_i\rangle = 0 \,\forall\, i \neq k\right\} \end{aligned}$$

We can immediately see that $\Omega \supseteq \mathbf{0}$. Consider the following

$$\begin{aligned} \Phi &= \sum_i w_i \cdot D_{\delta y}F_i\left(\delta y, 0\right)[w] \\ &= \sum_i w_i \cdot \sum_{k\in N_i} \langle w_k - w_i, y_k - y_i\rangle \cdot \alpha_{ik}\left(y_k - y_i\right) \\ &= \sum_{k,i\in N_{k,i}} \alpha_{ik}\langle w_k - w_i, y_k - y_i\rangle\langle w_i, y_k - y_i\rangle + \alpha_{ki}\langle w_i - w_k, y_i - y_k\rangle\langle w_k, y_i - y_k\rangle \\ &= -\sum_{k,i\in N_{k,i}} \alpha_{ik}\left(\langle w_k - w_i, y_k - y_i\rangle\right)^2 \end{aligned}$$

It follows that

$$w \in \Omega \Longrightarrow \Phi \equiv \mathbf{0}$$

and

$$\alpha_{ik} \neq 0 \Longrightarrow \langle w_k - w_i, y_k - y_i \rangle = 0 \, \forall \, i \neq k$$

which is precisely the defining condition placed on Ω. Thus $\{(w_1, w_2, w_3) \,|\, \langle w_k - w_i, y_k - y_i \rangle = 0 \, \forall \, i \neq k\} \subseteq \Omega$, therefore (4.4) has a nontrivial kernel. For example, take all three vectors equal. (Let $v \in \mathbb{R}^2$ be any nonzero vector and set

$$w_1 = w_2 = w_3 = v.$$

Then $w_k - w_i = 0$ for every pair (i, k), hence $\langle w_k - w_i, y_k - y_i \rangle = 0$. Therefore $(v, v, v) \in \Omega$. Because v can be chosen arbitrarily in $\mathbb{R}^2$, the kernel Ω contains at least the $2-$dimensional subspace

$$\left\{(v, v, v) \mid v \in \mathbb{R}^2\right\} \subsetneq \{0\},$$

which is nontrivial. In particular, $\Omega \neq \{0\}$.) $\square$

Assume $(w_1, w_2, w_3) \in \Omega$. Consider

$$\begin{aligned}
P_w F(\delta y, R) &= \sum_i w_i \cdot F_i(\delta y, R) \\
&= \sum_i \left[\langle w_i, H_V y_i \rangle + \left\langle w_i, H_V \frac{\delta y_i}{\tilde{m}_R} \right\rangle - \sum_{k \in N_i} \left[\langle w_i, \alpha_{ik}(R)(y_k - y_i) \rangle + \alpha_{ik}(R) \left\langle w_i, \frac{\delta y_k - \delta y_i}{\tilde{m}_R} \right\rangle \right] \right] \\
&= \frac{1}{\tilde{m}_R} \sum_i \left[\langle w_i, H_V \delta y_i \rangle - \sum_{k \in N_i} \alpha_{ik}(R) \langle w_i, \delta y_k - \delta y_i \rangle \right] \\
&= \frac{1}{\tilde{m}_R} \sum_i \left\langle w_i, H_V \delta y_i - \sum_{k \in N_i} \alpha_{ik}(R)(\delta y_k - \delta y_i) \right\rangle
\end{aligned}$$

we then multiply by $\tilde{m}_R$ and extend continuously to $R \to 0$ via

$$P_w F(\delta y, 0) = \sum_i \left\langle w_i, H_V \delta y_i - \sum_{k \in N_i} \alpha_{ik}(0)(\delta y_k - \delta y_i) \right\rangle$$

with $\alpha_{ik}(0) = e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle} \chi$. Now consider $P_\perp F\left(\delta y^0, 0\right)$ and $P_\parallel F\left(\delta y^0, 0\right)$ where $\perp, \parallel$ are with respect to Ω. We then have

$$\begin{aligned}
D_{\delta y} P_\perp F\left(\delta y^0, 0\right)[\delta w] &= P_\perp \sum_{k \in N_i} \langle \delta w_k - \delta w_i, y_k - y_i \rangle \alpha_{ik}(0)(y_k - y_i) \\
D_{\delta y} P_\parallel F\left(\delta y^0, 0\right)[\delta w] &= \sum_i \langle w_i, H_V \delta w_i \rangle \\
&\quad + \sum_i \sum_{k \in N_i} \left[\left\langle w_i, \alpha_{ki}(0)\left(\delta y_k^0 - \delta y_i^0\right) \right\rangle \langle y_k - y_i, \delta w_k - \delta w_i \rangle - \alpha_{ki}(0) \langle w_i, \delta w_k - \delta w_i \rangle \right]
\end{aligned}$$

For $\delta w \in \Gamma = \ker\left(D_{\delta y} P_\perp F\left(\delta y^0, 0\right) \oplus D_{\delta y} P_\parallel F\left(\delta y^0, 0\right)\right) \Rightarrow \delta w \in \Omega$ and

$$\sum_i \left\langle w_i, H_V \delta w_i - \sum_{k \in N_i} \alpha_{ki}(0) \langle w_i, \delta w_k - \delta w_i \rangle \right\rangle = 0$$

for all $w \in \Gamma$. Alternatively, $\delta w \in \Gamma$ if and only if $\delta w \in \Omega$ and

$$\begin{aligned} P_{\delta w} &= H_V \delta w_i - \sum_{k \in N_i} \alpha_{ki}(0)(\delta w_k - \delta w_i) \\ \Omega^{\perp} &\ni \\ &= \left\{ z \in \mathbb{R}^{2\times 3} | \exists c_{ki} = c_{ik} | z_i = \sum_{k \in N_i} c_{ki}(y_k - y_i) \right\} \end{aligned}$$

Claim 5. If we restrict δy_1 to be orthogonal to $\boldsymbol{r_1}$ and $\delta y_3 - \delta y_2$ to be parallel to $\boldsymbol{r_1}$. From $\sum_i F_i(\delta y, R) = 0$ we can find $\delta y_1(R), \delta y_2(R), \delta y_3(R)$ that solve (4.1) and $\delta y_1(0) = \delta y_1^0, \delta y_2(0) = \delta y_2^0, \delta y_3(0) = \delta y_3^0$.

Proof. Starting with $\sum_i F_i(\delta y, R) = 0$

$$\begin{aligned} \sum_i F_i(\delta y, R) &= 0 \\ \sum_i H_V y_i + H_V \frac{\delta y_i}{\tilde{m}_R} - \sum_{k \in N_i} \frac{2Q_R(|\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|)}{\|u_R\|_{L^2}^2 R^4 |\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|} \left(y_k - y_i + \frac{\delta y_k - \delta y_i}{\tilde{m}_R} \right) &= \\ \sum_i H_V \left(\frac{\tilde{m}_R y_i + \delta y_i}{\tilde{m}_R} \right) - \sum_{k \in N_i} \frac{2Q_R(|\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|)}{\|u_R\|_{L^2}^2 R^4 |\tilde{m}_R(y_k - y_i) + \delta y_k - \delta y_i|} \left(\frac{\tilde{m}_R y_k - \tilde{m}_R y_i + \delta y_k - \delta y_i}{\tilde{m}_R} \right) &= \\ \sum_i H_V \frac{\tilde{y}_i}{\tilde{m}_R} - \sum_{k \in N_i} \frac{2Q_R(|\tilde{m}_R y_k + \delta y_k - (\tilde{m}_R y_i + \delta y_i)|)}{\|u_R\|_{L^2}^2 R^4 |\tilde{m}_R y_k + \delta y_k - (\tilde{m}_R y_i + \delta y_i)|} \left(\frac{\tilde{m}_R y_k + \delta y_k - (\tilde{m}_R y_i + \delta y_i)}{\tilde{m}_R} \right) &= \\ \sum_i H_V \frac{\tilde{y}_i}{\tilde{m}_R} - \sum_{k \in N_i} \frac{2Q_R(|\tilde{y}_k - \tilde{y}_i|)}{\|u_R\|_{L^2}^2 R^4 |\tilde{y}_k - \tilde{y}_i|} \cdot \frac{\tilde{y}_k - \tilde{y}_i}{\tilde{m}_R} &= \\ &\Downarrow \\ H_V \frac{\tilde{y}_i}{\tilde{m}_R} - \sum_{k \in N_i} \frac{2Q_R(|\tilde{y}_k - \tilde{y}_i|)}{\|u_R\|_{L^2}^2 R^4 |\tilde{y}_k - \tilde{y}_i|} \cdot \frac{\tilde{y}_k - \tilde{y}_i}{\tilde{m}_R} &= 0 \end{aligned}$$

which is precisely (4.1), and since

$$\begin{aligned} H_V \frac{\tilde{y}_i}{\tilde{m}_R} &= \sum_{k \in N_i} \frac{2Q_R(|\tilde{y}_k - \tilde{y}_i|)}{\|u_R\|_{L^2}^2 R^4 |\tilde{y}_k - \tilde{y}_i|} \cdot \frac{\tilde{y}_k - \tilde{y}_i}{\tilde{m}_R} \\ &\Downarrow (R \searrow 0) \\ H_V y_i &= \sum_{k \in N_i} \alpha_{ki} \chi(y_k - y_i) \end{aligned}$$

has solutions (by construction) δy_i^0, we also can see $\delta y_i(0) = \lim_{R \searrow 0} \delta y_i(R) = \delta y_i^0$. □

More generally, $(\delta y_1, \delta y_2, \delta y_3) \perp (w_1, w_2, w_3) = w_{\text{rot}} \iff \delta y \in \{w_{\text{rot}}\}^{\perp}$. $F_i(\delta y_i, R)$ does not necessarily preserve $\{w_{\text{rot}}\}^{\perp}$; however, $\{w_{\text{rot}}\}^{\perp} = P_{\perp_{\text{rot}}} F_i(\delta y_i, R) = 0$ has a unique solution. If $F_i(R_w \delta y_i, R) = R_w F_i(\delta y_i, R)$ for all $w \in [-\pi, \pi]$, then $F_i(R_w \delta y_i, R) = 0$ such that $F_i(\delta y_i, R) \mapsto R_w F_i(\delta y_i, R)$ (since $F_i(\delta y_i, R) = 0$).

5 Conclusion

We have carried out a complete finite-dimensional study of multi-peak ground states for the nonlinear Schrödinger equation

$$(-\Delta + V + E)\psi + \sigma|\psi|^{2p}\psi = 0, \qquad E \to \infty,$$

using the rescaling and concentration compactness framework. The existence question is reduced to an algebraic system

$$F_i(\delta y, R) = 0,$$

which governs the positions of well-separated profiles near a non-degenerate critical point of the potential V.

Our main results are threefold.

1. Classification of limiting configurations ($R = 0$).
   (a) Collinear peaks along a single Hessian eigenvector.
   (b) Isosceles and equilateral triangles in the plane spanned by two eigenvectors.
   (c) Equilateral triangles with a continuous rotational degree of freedom θ. For the rotational case we derived admissible regions in the $(\mu, \theta)-$plane, where $\mu = \frac{|\lambda_2|}{|\lambda_1|}$. All orientations are admissible when $\frac{1}{3} \leq \mu \leq 3$; otherwise only restricted angular windows remain.
2. Perturbation of the limiting system ($R > 0$).
   (a) We formulated the system $F_i(\delta y, R) = 0$ and constructed explicit solutions δy^0 at $R = 0$ for the equilateral triangle (Claim 3).
   (b) The linearization $D_{\delta y} F_i(\delta y^0, 0)$ has a nontrivial kernel Ω containing the two-dimensional subspace of rigid translations, which blocks a direct application of the implicit function theorem.
3. Resolution strategy for the kernel obstruction.
   (a) Following the iterative kernel-decomposition method, we decomposed the reduced system into components perpendicular and parallel to the kernel.
   (b) Gauge-fixing constraints (removing the rotational freedom) yield a projected system that is expected to admit a unique branch, provided the full system is rotationally equivariant.

The work stops at a precise point: the concentration compactness decomposition, the Lyapunov-Schmidt reduction, and the link back to the full PDE are already established. What remains is to finish the finite-dimensional step by:

1. Verifying rotational equivariance of F_i.
2. Proving invertibility of the linearized operator on the gauge-fixed subspace.
3. Extending the analysis to configurations with a continuous rotation angle θ.

Once these are done, the Lyapunov-Schmidt machinery lifts the finite-dimensional solutions to exact multi-peak ground states for all sufficiently large energies E. Stability properties are largely understood: multi-profile branches with two or more peaks have at least two negative directions in the linearized operator, implying orbital instability; asymptotic stability of single-peak states has been established in earlier works, and adapting those techniques to multi-peak states remains an open problem.

In summary, we have classified all admissible multi-peak geometries, identified the kernel obstruction, and set up the finite-dimensional reduction. The remaining tasks (equivariance, invertibility, and handling of the $\theta-$dependent case) are concrete, finite-dimensional problems whose solution will complete the existence theory for multi-peak ground states at finite energy.

6 Notation

6.1 Equation and Basic Parameters

Symbol	Description
$\psi : \mathbb{R}^n \to \mathbb{C}$	Solution of the nonlinear Schrödinger equation
$V : \mathbb{R}^n \to \mathbb{R}$	External potential (encodes refractive index profile)
$E \in \mathbb{R}^+ \cup \{0\}$	Energy parameter (propagation constant in optics)
$\sigma \in \mathbb{R}^+$	Nonlinearity coefficient (sign determines focusing/defocusing)
p	Nonlinear power exponent; $p = 1$ corresponds to cubic (Kerr) nonlinearity
x_0	Non-degenerate critical point of V, i.e., $\nabla V(x_0) = 0$ with $H_V(x_0)$ invertible

Table 6.1: Notation - Equation and Basic Parameters

6.2 Scaling and Profile Functions

Symbol	Description
$R = E^{-\frac{1}{2}}$	Scaling parameter; $R \to 0$ as $E \to \infty$
$u_E(x) = E^{-\frac{1}{2p}} \psi_E\left(E^{-\frac{1}{2}} x\right)$	Rescaled solution with $\|u_E\|_{H^1}$ uniformly bounded
u_R	Profile function solving $\left(-\Delta + R^2 V(x_0) + 1\right) u + \sigma \|u\|^{2p} u = 0$
$M \in \mathbb{N}$	Number of peaks in the multi-peak decomposition
$z_i(R)$	Position of the $i-$th peak center in rescaled coordinates
h_R	Remainder term: $u_E = \sum_{i=1}^{M} u_R(\cdot - z_i) + h_R$

Table 6.2: Notation - Scaling and Profile Functions

6.3 Hessian and Spectral Properties

Symbol	Description
$H_V = H_V(x_0)$	Hessian matrix of V at the critical point x_0
λ_i	Eigenvalue of H_V; assumed $\lambda_i < 0$ for directions in which peaks spread
$\boldsymbol{r}_i$	Unit eigenvector of H_V associated with eigenvalue λ_i
$\mu = \frac{\|\lambda_2\|}{\|\lambda_1\|}$	Eigenvalue ratio; key parameter for admissibility constraints
U	Unstable manifold: $U = \operatorname{span}\{\boldsymbol{r} : H_V \boldsymbol{r} = \lambda \boldsymbol{r}, \lambda < 0\}$

Table 6.3: Notation - Hessian and Spectral Properties

6.4 Bump Positions and Geometry

Symbol	Description
$s_i = Rz_i - x_0$	Displacement of $i-$th peak from critical point x_0
m_R	Minimum inter-peak distance: $m_R = \min_{i \neq j} \lvert z_i(R) - z_j(R) \rvert$
$y_i = \lim_{R \to 0} \frac{s_i}{Rm_R}$	Limiting position vector of the $i-$th peak; satisfies $\lvert y_k - y_i \rvert = 1$ for nearest neighbors
N_i	Set of nearest neighbors of peak i: $N_i = \{k : \lvert y_k - y_i \rvert = 1\}$
a, h	Geometric parameters for triangular configurations: a = half-base, h = apex height

Table 6.4: Notation - Bump Positions and Geometry

6.5 Interaction Functional and Coefficients

Symbol	Description
$Q_R(z)$	Interaction functional: $Q_R(z) = -\sigma \left\langle \partial_{\frac{z}{\lvert z \rvert}} u_R(\cdot - z), u_R^{2p+1} \right\rangle$
$\alpha_{ki} = \frac{Q_R(\lvert z_k - z_i \rvert)}{Q_R(m_R)}$	Interaction coefficient between peaks k and i; satisfies $0 \leq \alpha_{ki} \leq 1$
$\tilde{\alpha}_{ki} = e^{-\langle y_k - y_i, \delta y_k - \delta y_i \rangle}$	Perturbed interaction coefficient at $R = 0$
$\chi = \frac{2Q_R(m_R)}{R^4 m_R}$	Overall interaction strength; $\chi > 0$

Table 6.5: Notation - Interaction Functional and Coefficients

6.6 Reduced System and Perturbations

Symbol	Description
$\tilde{y}_i = \tilde{m}_R y_i + \delta y_i$	Perturbed peak position; $\frac{\delta y_i}{\tilde{m}_R} \to 0$ as $R \to 0$
$\delta y_i \in \operatorname{span}\{\boldsymbol{r}_1, \boldsymbol{r}_2\}$	Perturbation of the $i-$th peak position
$\delta y^0 = \left(\delta y_1^0, \delta y_2^0, \delta y_3^0\right)$	Explicit solution at $R = 0$ with $\delta y_1^0 = -\gamma \boldsymbol{r}_2$, $\delta y_{2,3}^0 = \mp \beta \boldsymbol{r}_1 + \frac{\gamma}{2} \boldsymbol{r}_2$
$\beta = \frac{1}{2} \ln \left(\frac{\alpha_{12}}{\alpha_{23}} \right)$	Parameter in the explicit solution δy^0
$\gamma = \frac{2\beta}{3\sqrt{3}}$	Parameter in the explicit solution δy^0
$F_i(\delta y, R)$	Perturbation system governing peak positions. $F_i(\delta y, 0) = 0$ characterizes limiting configurations

Table 6.6: Notation - Reduced System and Perturbations

6.7 Kernel and Projections

Symbol	Description
Ω	Kernel of $D_{\delta y}F_i\left(\delta y^0, 0\right)$; contains $\left\{(v,v,v) : v \in \mathbb{R}^2\right\}$ (translations)
$\Omega^{\perp}$	Orthogonal complement: perturbations expressible as $z_i = \sum_{k\in N_i} c_{ki}\left(y_k - y_i\right)$
$P_{\perp}F$, $P_{\parallel}F$	Projections of F perpendicular and parallel to the kernel Ω
$P_w F\left(\delta y, R\right)$	Projection along $w \in \Omega$: $P_w F = \sum_i \langle w_i, F_i\rangle$
Γ	Joint kernel: $\Gamma = \ker\left(D_{\delta y}P_{\perp}F \oplus D_{\delta y}P_{\parallel}F\right)$
$P_{\delta w}$	Linear operator acting on perturbations δw_i: $P_{\delta w} = H_V \delta w_i - \sum_{k\in N_i} \alpha_{ki}\left(0\right)\left(\delta w_k - \delta w_i\right)$
w_{rot}	Generator of the rotational kernel direction

Table 6.7: Notation - Kernel and Projections

6.8 Rotations and Equivariance

Symbol	Description
θ	Rotation angle; orientation of triangular configuration relative to eigenvector basis
$R\left(\theta\right)$	2D rotation matrix: $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
$R^{-1}\left(\theta\right) = R\left(-\theta\right)$	Inverse rotation matrix: $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$
R_{ω}	Rotation by angle ω. Used in equivariance condition $F_i\left(R_{\omega}\delta y, R\right) = R_{\omega}F_i\left(\delta y, R\right)$
$y_i' = R^{-1}\left(\theta\right) y_i$	Bump positions in the rotated (canonical) frame

Table 6.8: Notation - Rotations and Equivariance

6.9 Admissibility Functions

Symbol	Description
$\nu\left(\mu\right) = \frac{\mu+1}{2\lvert\mu-1\rvert}$	Auxiliary function for admissibility. Determines critical angle
$\varphi\left(\mu\right) = \arccos\left(\nu\left(\mu\right)\right)$	Critical angle function: $\varphi = \frac{\pi}{3}$ when $\nu \leq \frac{1}{2}$, $\varphi = 0$ when $\nu \geq 1$
(μ, θ) −space	Parameter space for rotational equilateral configurations

Table 6.9: Notation - Admissibility Functions

6.10 Inner Products and Norms

Symbol	Description
$\langle \cdot, \cdot \rangle$	Standard Euclidean inner product in $\mathbb{R}^n$ or L^2 inner product (context-dependent)
$\lvert \cdot \rvert$	Euclidean norm in $\mathbb{R}^n$
$\lVert \cdot \rVert_{H^1}$	Sobolev H^1 norm: $\lVert u \rVert_{H^1}^2 = \lVert \nabla u \rVert_{L^2}^2 + \lVert u \rVert_{L^2}^2$
$\lVert \cdot \rVert_{L^2}$	Standard L^2 norm
$\perp$	Orthogonality (with respect to L^2 or Euclidean inner product)

Table 6.10: Notation - Inner Products and Norms

6.11 Linearization Operators

Symbol	Description
$D_{\delta y} F_i$	Fréchet derivative of F_i with respect to δy
$D_{\delta y} F_i \left(\delta y^0, 0\right) [w]$	Linearization at $\left(\delta y^0, 0\right)$ applied to direction w
L_+	Linearized operator at the Lyapunov-Schmidt level. Kernel generated by translational invariance
$D_\psi F \left(\psi_E, E\right)$	Linearization of the full PDE at a solution ψ_E

Table 6.11: Notation - Linearization Operators

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